Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems

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Presentation Outline

The Rise of AI in Cyber-Physical Systems (CPS)

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The Catastrophic Vulnerability: The "Black Box"

Despite high accuracy, AI models act as opaque "black boxes". In safety-critical environments, when an AI flags an anomaly, human operators cannot answer: **"Why did the AI make this decision?"**

Literature Survey

Reference (IEEE Format)	Core Focus	Identified Gap / Limitation
Wickramasinghe et al. (2021) [1]	Explainable unsupervised ML for cyber-physical systems.	**Context-Agnostic:** Evaluates sensors in a vacuum without external environmental factors.
Ahmed et al. (2022) [2]	XAI applications in Industry 4.0 and smart manufacturing.	**Evaluation Gap:** Focuses on algorithmic accuracy; lacks formalized, human-centered evaluation standards.
Islam et al. (2021) [3]	Comprehensive survey of XAI approaches and metrics.	**Metric Flaw:** Evaluation metrics do not measure operational "Actionability" or "Trust" in high-stakes CPS.
Sofianidis et al. (2021) [4]	Review of ethical and human-centric AI in smart production.	**Implementation:** Purely theoretical framework; lacks empirical, multi-agent software implementation.

The Crisis of Traditional Explainable AI (XAI)

To mitigate the "black box" crisis, Explainable AI (XAI) frameworks like SHAP and LIME have become industry standards. However, they suffer from a severe operational limitation.

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Operational Consequence

Context-agnostic explanations generate costly false positives, misdiagnose environmental anomalies as physical hardware failures, and rapidly degrade operator trust.

Project Objectives & The "Glass Box" Approach

Primary Goal: To transition from theoretical XAI models to a functional, enterprise-ready software prototype that actively understands its environment.

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4. **Automate Human-Centered Evaluation:** Pioneer an "LLM-as-a-Judge" pipeline to objectively score explanations, eliminating the bottleneck of manual human testing.

Redefining Metrics: Why Trust, Reasonableness, Actionability?

The Problem with Standard Metrics

Metrics like Accuracy or F1-scores measure if a model is *correct*, but not if the explanation is *useful*. Traditional XAI (like SHAP) provides numerical feature weights, not human-readable reasoning.

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- **Reasonableness ("Does this make logical sense?"):** Checks if the AI's logic aligns with established domain physics and expert engineering knowledge.
- **Actionability ("What do I actually do now?"):** An explanation is useless if it doesn't guide a concrete next step under time pressure.

The Data Challenge in Safety-Critical CPS

The Invigilator Question: Why not use real-world data?

While real-world telemetry is the gold standard for testing AI, accessing raw, labeled data from modern Cyber-Physical Systems is practically impossible for two critical reasons:

- **National Security & Air-Gapping:** Smart Power Grids and Water Treatment Plants are critical infrastructure. Their SCADA network telemetry is strictly proprietary, classified, and air-gapped to prevent cyber-terrorism.

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- **The Contextual Labeling Deficit:** Even if real data is obtained, it only records the *internal* failure (the "What"). Real datasets rarely have perfectly synchronized labels for *external* events (weather, latency) which are mandatory for testing Context-Aware XAI.

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Our Solution: Synthesize a physics-grounded, dual-domain dataset that intentionally isolates mechanical failures from environmental anomalies to rigorously stress-test the XAI's reasoning.

Dataset Description & Source

To ensure high operational realism, we engineered two 1,000-sample time-series datasets (15-minute intervals) using Gaussian distributions centered on real engineering standards.

Smart Power Grid

- **Internal Base:** 132kV Voltage, 400A Current, 50Hz Frequency.
- **Contextual Anomalies:** Heatwaves causing thermal stress; Lightning causing latency.
- **Mechanical Failures:** Isolated Transformer/Line faults.

Smart Water Plant

- **Internal Base:** 50 psi Pressure, 2.0 mm/s Pump Vibration.
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Ensuring Legitimacy: Our Python generation script binds all normal states within strict tolerances (e.g., Grid frequency $\approx 49.9 - 50.1$ Hz). When anomalies are injected, the telemetry mathematically deviates in ways that accurately mimic physical engineering laws.

Operational Workflow: XAI-CPS Ecosystem

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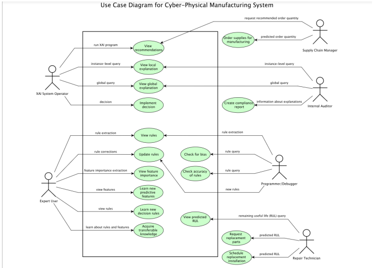

FIGURE 4. Use case diagram demonstrating implementation of XAI for a cyber-physical manufacturing system.

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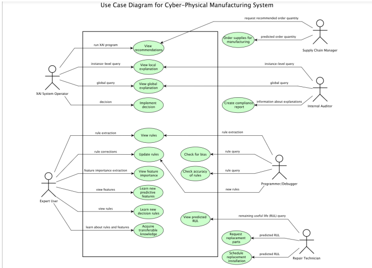

FIGURE 4. **Use case diagram demonstrating implementation of XAI for a cyber-physical manufacturing system.**

- **Persona-Driven Explainability:** An Operator needs immediate actionability, while an Auditor needs global rule extraction. Our framework is designed to serve this diverse ecosystem.

Data Preprocessing & Anomaly Filtering

Before the raw dataset is passed to our XAI Multi-Agent pipeline, it must undergo strict preprocessing and rule-based threshold filtering.

Preprocessing Steps:

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3. **LLM Triggering:** Computationally heavy LLM inference is completely paused until this preprocessing step flags a critical event, ensuring real-time system efficiency.

System Architecture: Step 2 - Multi-Agent Orchestration

Tech Stack: Microsoft AutoGen + LLaMA 3.2 (via Ollama, 100% Offline).

Once an anomaly is flagged, it is passed to a three-agent ecosystem:

- **Agent 1a (Context-Agnostic Explainer):** Represents traditional XAI. Receives *only* internal sensor data. Assumes mechanical failure by default.

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Note: By making the causal reasoning process explicit and transparent via LLM prompting, we move beyond "black-box" scores to true, human-understandable AI.

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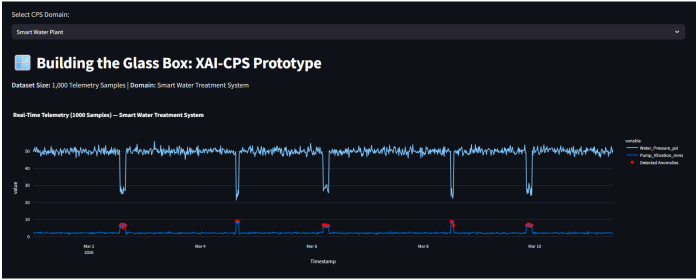

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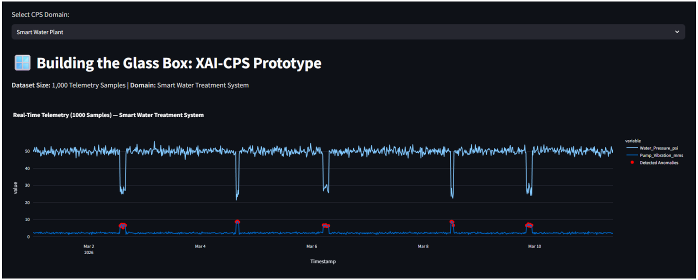

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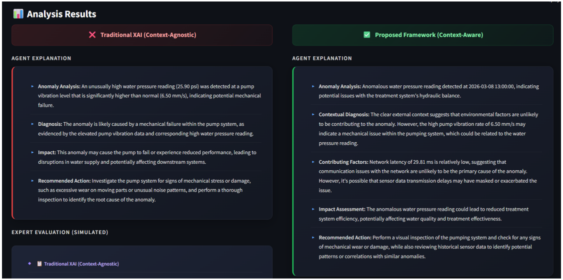

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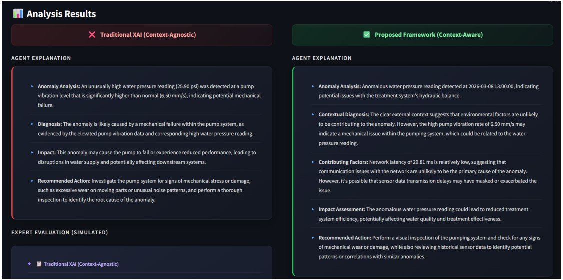

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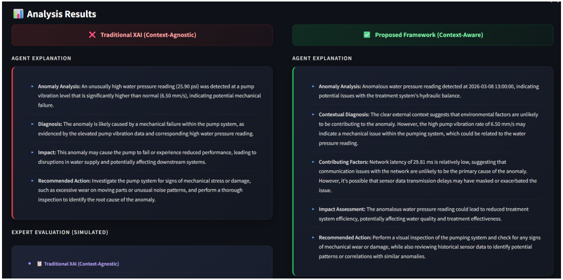

- **Traditional XAI (Left):** Diagnoses a broken pump/transformer purely based on pressure/voltage drops. Recommends costly physical repair.
- **Proposed Framework (Right):** Ingests external variables to realize the grid is overworking to compensate for a heatwave/storm.

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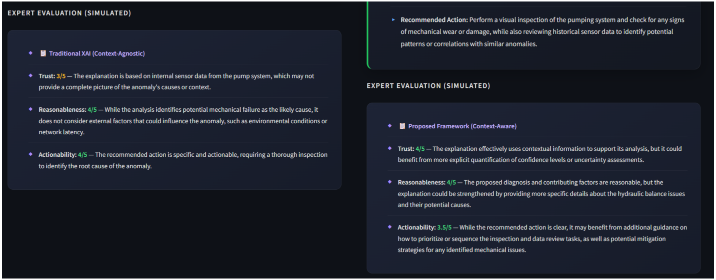

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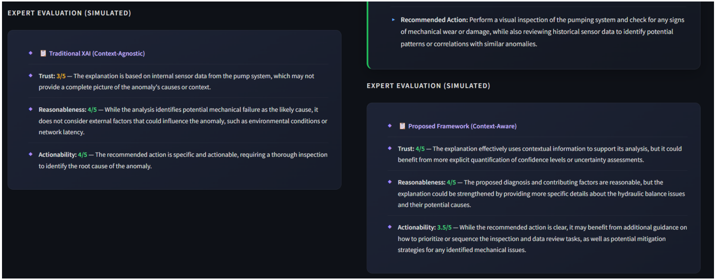

- **The Result:** Our proposed context-aware model consistently achieves better scores in actionability, Reasonableness, and Trust by eliminating context-agnostic false positives.

Conclusion & Future Research Directions

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3. **Biometric Validation:** Validating our automated LLM-as-a-Judge scores against human eye-tracking or EEG scans.

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Thank You

Questions & Discussion